Event Cameras for Non-Line-of-Sight Optical Communication: A Survey of Technical Evolution and Future Directions

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September 2026

Abstract

Event cameras—also termed neuromorphic vision sensors or dynamic vision sensors (DVS)—have emerged as a transformative receiver technology for optical camera communication (OCC), addressing the temporal sampling bottleneck that restricts conventional frame-based cameras to sub-kHz effective bandwidths. By asynchronously reporting pixel-level log-intensity changes with microsecond temporal resolution and a dynamic range exceeding 120 dB, event cameras support communication links that are simultaneously high-rate, motion-tolerant, and robust to extreme illumination conditions—attributes of particular value when the line-of-sight (LOS) path is obstructed and the receiver must recover data from diffuse reflections. This survey traces the technical evolution of event-camera-based optical communication from its hardware origins (2008–2017) through four developmental stages: proof-of-concept event-OCC (2017–2018), spatial encoding and beacon systems (2022–2024), the first non-line-of-sight (NLOS) event-camera demonstrations (2025), and long-range outdoor together with specialized deployments (2025–2026). The literature is organized along three axes: hardware capability, modulation and demodulation design, and channel environment (LOS, NLOS, and underwater). For each stage, the key technical mechanisms are delineated—including On–Off Keying (OOK), N-pulse modulation, event interval modulation, toggle demodulation, 8b/10b-assisted clock recovery, dynamic QR-code spatial encoding, and adaptive reflection-based NLOS schemes—and the experimentally demonstrated data-rate–versus–distance frontier is tabulated. The survey concludes with a discussion of open challenges and a research roadmap toward event-camera-enabled NLOS communication networks for sixth-generation (6G) applications.

Index Terms—Event camera, neuromorphic vision sensor, dynamic vision sensor (DVS), optical camera communication (OCC), non-line-of-sight (NLOS), visible light communication (VLC), rolling shutter, modulation, clock recovery, 6G.

1 Introduction

Optical Camera Communication (OCC) employs an image sensor as the receiver for visible light communication (VLC), leveraging the camera's spatial resolution to discriminate among multiple LED transmitters and suppress ambient interference. The principal limitation of conventional frame-based OCC is temporal: a commodity camera acquires 30–60 frames per second, yielding an effective sampling bandwidth below 1 kHz even when rolling-shutter stripe extraction is exploited. LED transmitters, by contrast, are readily modulated in the MHz range. This sampling mismatch—spanning three to four orders of magnitude—imposes a fundamental ceiling on frame-based OCC throughput and renders the system acutely susceptible to motion blur and frame-synchronization errors.

Event cameras, alternatively referred to as Dynamic Vision Sensors (DVS) or neuromorphic vision sensors, overcome this bottleneck through a fundamentally different sensing paradigm. Each pixel operates asynchronously and independently, emitting an *event* (x, y, t, p) only when the logarithmic intensity change exceeds a preset threshold, where t denotes a microsecond-resolution timestamp and $p \in \{+1, -1\}$ indicates the polarity (brightness increase or decrease). The resulting sparse, asynchronous event stream exhibits three properties that are particularly advantageous for OCC: (i) **high temporal resolution** ($\sim$1 μs, corresponding to an effective sampling rate of $\sim$20–1000 kHz), (ii) **wide dynamic range** (>120 dB, compared with $\sim$60–70 dB for conventional CMOS sensors), and (iii) **low data volume and latency** (only pixels exhibiting intensity changes transmit data, thereby eliminating redundant background frames).

These properties assume heightened significance in Non-Line-of-Sight (NLOS) OCC, wherein the direct LED-to-camera optical path is obstructed and the receiver must recover data from diffuse reflections off walls, the ground, or arbitrary objects. The reflected signal undergoes severe attenuation, temporal smearing by the bidirectional reflectance distribution function (BRDF) of the reflecting surface, and superposition with

spatially heterogeneous background illumination. The event camera's wide dynamic range enables detection of the faint reflected intensity variations that a frame-based sensor would embed in quantization noise, while its asynchronous architecture eliminates the frame-synchronization impediment that afflicts reflected signals subject to unknown and geometry-dependent propagation delays.

Scope and contributions. This survey focuses specifically on event cameras as receivers for optical communication, with particular emphasis on the progression toward NLOS operation. The technical evolution is traced through four developmental stages (Section 3); the modulation and demodulation design space is analyzed in Section 4; the NLOS event-camera channel is examined in detail in Section 5; and open challenges together with a research roadmap are discussed in Section 6. The experimentally demonstrated data-rate–versus–distance frontier is summarized in Table 1, and the principal technical transitions underpinning each advance are identified throughout.

2 Hardware Foundations

The evolution of event-camera hardware has been a direct enabler of the communication performance gains documented in this survey. Three hardware milestones are of foundational significance:

DVS128 (2008). The first commercially available event camera, introduced by Lichtsteiner *et al.* [6], comprised a 128×128 pixel array with $40\,\mu$m pixel pitch, 120 dB dynamic range, and $15\,\mu$s latency. Although the 350 nm CMOS process yielded relatively high dark current, the asynchronous pixel architecture—a continuous-time logarithmic photoreceptor coupled to a self-timed threshold comparator—established the event-generation model inherited by all subsequent sensors. The modest resolution and elevated noise floor of the DVS128 constrained early event-OCC experiments to short ranges and low data rates.

DAVIS240 (2014). Brandli *et al.* [3] introduced the DAVIS240, a 240×180 sensor offering 130 dB dynamic range and $3\,\mu$s latency. A distinguishing feature of the DAVIS240 is its interleaving of event data with global-shutter intensity frames, thereby enabling hybrid processing pipelines that leverage frames for transmitter localization and events for high-speed demodulation.

Samsung VGA DVS (2017). Son *et al.* [12] demonstrated a 640×480 VGA-resolution DVS with $9\,\mu$m pixel pitch and 300 Meps (million events per second) aggregate throughput, fabricated using a back-side-illuminated (BSI) CMOS image sensor process. The enhanced spatial resolution and event throughput directly facilitated longer-range event-OCC links, as the receiver could resolve smaller LED images at greater distances while sustaining adequate event density for reliable demodulation. Subsequent sensor generations—exemplified by the Prophesee EVK4 integrating the Sony IMX636 (1280×720, $4.86\,\mu$m pixel)—have extended this trajectory, approaching the $\sim$1 MHz per-pixel event-rate ceiling exploited by the most recent long-range demonstrations [9].

3 Technical Evolution: Four Stages

The application of event cameras to optical communication has advanced through four identifiable stages, each demarcated by a principal technical transition. Table 1 summarizes the experimentally demonstrated performance milestones across all stages.

3.1 Stage 1: Proof-of-Concept Event-OCC (2017–2018)

The initial demonstrations of event cameras as OCC receivers appeared in 2017–2018. Perez-Ramirez *et al.* [10] established a 500 bit/s link over 5 m with BER $< 10^{-4}$ using On–Off Keying (OOK) and a DVS128 receiver. The underlying principle is that an OOK-modulated LED produces a rising-edge (ON) event at the pixels imaging the LED upon illumination onset and a falling-edge (OFF) event upon extinction; the event stream thus encodes the bit sequence directly, obviating the rolling-shutter stripe extraction indispensable to frame-based OCC.

Shen *et al.* [11] extended this paradigm to multi-level modulation, achieving 16 kbit/s at 10 m through inverse Pulse Position Modulation (i-PPM) combined with Frequency Shift Keying (FSK). This work was the first to exploit the event camera's microsecond temporal resolution for frequency-domain encoding, advancing beyond binary OOK. However, the bit error rate was not rigorously evaluated, and the system presupposed a static, aligned link; neither motion tolerance nor NLOS capability was investigated.

Table 1: Experimentally demonstrated event-OCC performance frontier across all developmental stages.

Year	Stage	Data Rate	Distance	BER	Key Technique
2017	**Stage 1**	500 bit/s	5 m	$< 10^{-4}$	OOK + DVS128 receiver [10]
2018	**Stage 1**	16 kbit/s	10 m	N/A	i-PPM + FSK multi-level [11]
2022	**Stage 2**	500 bit/s	100 m	0 (error-free)	Smart visual beacon + LED [17]
2024	**Stage 2**	2.5 kbit/s	0.5 m	94% acc.	Dynamic environment framework [16]
2024	**Stage 2**	—	—	—	N-pulse modulation [1]
2025	**Stage 3**	114 kbit/s	1.25 m	1%	Dynamic QR spatial encoding [13]
2025	**Stage 3**	—	indoor	—	First NLOS event-camera OCC [7]
2025	**Stage 4**	60 kbit/s	200 m	$< 10^{-3}$	OOK + toggle + digital PLL [15]
2025	**Stage 4**	30 kbit/s	400 m	$< 10^{-3}$	(outdoor, same system) [15]
2025	**Stage 4**	28 kbit/s	10 m	—	Event Interval Modulation (EIM) [14]
2026	**Stage 4**	80 kbit/s	320 m	$< 10^{-3}$	8b/10b + OOK-NRZ + PLL, 120° FoV [9]
2026	**Stage 4**	240 kbit/s	320 m	$< 10^{-3}$	Multi-node (3 beams, same system) [9]

Stage 1 summary. These studies established that event cameras can demodulate optical signals at rates substantially exceeding the frame-rate limitation. Nevertheless, the demonstrated throughput ($\leq$16 kbit/s) and range ($\leq$10 m) remained modest, and all links were confined to line-of-sight, static configurations.

3.2 Stage 2: Spatial Encoding and Beacon Systems (2022–2024)

The second stage introduced spatial encoding—exploiting the event camera's two-dimensional pixel array as a communication resource rather than a single-pixel detector—and extended the operational range by an order of magnitude.

Wang *et al.* [17] proposed a "smart visual beacon" architecture at IROS 2022, integrating an LED transmitter with event-camera demodulation algorithms. The system achieved 100 m at 500 bit/s with zero reported bit errors, accompanied by a quantitative evaluation of the trade-offs among LED transmission rate, communication distance, and message accuracy. The beacon architecture was conceived for robotic applications—an event camera mounted on a robot demodulates a beacon affixed to a stationary fixture—and demonstrated that event-OCC can operate at ranges pertinent to indoor navigation and vehicle-to-everything (V2X) scenarios.

Aranda *et al.* [1] introduced N-pulse modulation, a scheme purpose-designed for the event camera's operational principles. Rather than encoding each bit as a single intensity transition, N-pulse modulation represents a symbol as a group of N pulses with a defined temporal pattern, with the encoding adapted to the bit composition of the packet. This approach significantly reduced demodulation complexity relative to OOK and PPM, as the receiver can exploit the temporal structure of the pulse group rather than relying on per-edge thresholding. The same group subsequently evaluated pulse-shaped waveform modulation [2], leveraging the sensor's high temporal resolution through shaped pulses that enhance noise resilience.

Von Arnim *et al.* [16] addressed the practical challenge of dynamic environments, in which ambient illumination and target motion vary during communication. Their framework attained 2.5 kbit/s with 94% message accuracy at 0.5 m. Although these figures are modest in absolute terms, the principal contribution lies in robustness to time-varying channel conditions rather than raw performance.

Stage 2 summary. Stage 2 expanded event-OCC from a single-pixel proof of concept to a spatially encoded, robot-relevant communication modality. The operational range was extended to 100 m; however, throughput remained below 10 kbit/s, and all links were still confined to line-of-sight.

3.3 Stage 3: NLOS Event-Camera Communication (2025)

The third stage—constituting the central focus of this survey—marks the first demonstrations of event cameras for NLOS communication, wherein the signal is recovered from reflections rather than from a direct line-of-sight path.

3.3.1 First NLOS event-camera OCC

Nishar *et al.* [7] (EWSN 2025, arXiv:2503.11226) reported the first NLOS optical wireless communication system employing a neuromorphic camera. The principal innovation lies in leveraging *passive reflections from everyday objects* as the communication channel: an LED modulated with OOK or N-pulse modulation

illuminates a surface (e.g., a wall, poster, or piece of furniture), and the event camera detects the intensity variations reflected from that surface—notwithstanding the absence of a direct view of the LED. The system overcomes a fundamental constraint of event cameras—namely, that intensity *changes* are required to generate events and a perfectly static scene produces none—by deliberately inducing VLC-modulated intensity variations of sufficient magnitude to generate events via reflection, even under zero-motion conditions.

The paper further introduced an *adaptive N-pulse modulation* scheme that dynamically adjusts the encoding in accordance with the bit composition of the packet, thereby optimizing the trade-off between data rate and robustness. Experimental results demonstrated that lighter-colored, glossy objects serve as superior NLOS communication surfaces, owing to higher reflectivity and a more specular reflection component. Conversely, larger objects and matte finishes exhibited elevated error rates attributable to multipath reflections that smear the temporal structure of the event stream.

3.3.2 Motion-aware spatial encoding

Su *et al.* [13] (IEEE RAL 2025, arXiv:2412.00816) introduced a complementary advance: a dynamic visual marker—effectively a QR-code-like spatial pattern rendered on a display—coupled with event-based tracking and decoding algorithms. The event camera simultaneously tracks the marker's corners and decodes the spatially encoded data, achieving 114 kbit/s under static conditions—the highest short-range event-OCC throughput reported to date—along with 1 cm localization accuracy at 1% BER under various camera motions. The contribution of particular relevance to NLOS is that the event camera's intrinsic immunity to motion blur preserves communication capability even when the receiver is in motion, a scenario that renders frame-based OCC unreliable due to blur-induced stripe distortion. The open-source code and accompanying dataset are publicly available at `https://github.com/suhang99/EventOCC`.

Stage 3 summary. Stage 3 established that event cameras can sustain communication via NLOS reflections and under receiver motion—two capabilities unattainable with frame-based cameras. Throughput reached 114 kbit/s (short-range, spatial encoding); however, NLOS-through-reflection throughput remained limited to the order of kbit/s, constrained by the attenuation and multipath characteristics of the reflection channel.

3.4 Stage 4: Long-Range and Specialized Deployments (2025–2026)

The fourth stage, presently unfolding, extends event-OCC to deployment-relevant ranges and environments, encompassing outdoor long-range links, underwater communication, and multi-node network configurations.

3.4.1 Long-range outdoor

Sumino *et al.* [15] (arXiv:2510.14266, GLOBECOM 2025) reported the first long-range outdoor event-OCC system achieving BER $< 10^{-3}$: 60 kbit/s at 200 m and 30 kbit/s at 400 m, employing OOK with toggle demodulation and a digital phase-locked loop (PLL) for clock recovery. The toggle demodulation scheme exploits the polarity output of the event camera: each OOK symbol generates an ON-event burst followed by an OFF-event burst, and the demodulator alternates its bit decision predicated on the event polarity, thereby providing intrinsic noise rejection. The digital PLL recovers the symbol clock from the event timestamps, accommodating the timing jitter introduced by atmospheric turbulence.

Nardo *et al.* [9] (arXiv:2609.29776) further advanced the state of the art, demonstrating 80 kbit/s at 320 m with a 120° field-of-view fisheye lens, utilizing 8b/10b line coding with OOK-NRZ and a Bang-Bang PLL. The 8b/10b coding ensures at least one edge transition every five symbols, guaranteeing continuous event generation even during extended runs of identical bits. In multi-node operation, three parallel beams yielded an aggregate throughput of 240 kbit/s, with real-time processing latency below 10 ms per 50 ms event batch. The system employed a Prophesee EVK4 (Sony IMX636) sensor; the authors identified the sensor readout bottleneck ($\sim$100 lines/μs) as the factor precluding attainment of the theoretical 1 MHz bandwidth in practice.

3.4.2 Novel modulation: Event Interval Modulation

Sumino *et al.* [14] (arXiv:2510.14245) proposed Event Interval Modulation (EIM), the first modulation scheme designed expressly for event cameras. Rather than encoding information in the presence or absence of events (OOK) or in pulse patterns (N-pulse), EIM encodes information in the *time intervals between consecutive events*. This approach fully exploits the asynchronous temporal resolution of the event camera and facilitates higher-order modulation: the interval duration maps to a symbol, such that M-ary EIM conveys $\log_2 M$ bits per event. The proof-of-concept demonstration achieved 28 kbit/s at 10 m and 8.4 kbit/s at 50 m indoors, establishing a new benchmark for event-OCC bit rate in the moderate-range regime.

3.4.3 Underwater event-camera ISAC

Fang *et al.* [4] (Frontiers in Marine Science, 2026) extended event-camera OCC to the underwater domain, presenting an event-based camera-enabled underwater optical integrated sensing and communication (ISAC) system. This work is noteworthy as the first cross-medium event-OCC demonstration: the wavelength-dependent absorption and particulate scattering characteristic of the underwater channel are analogous to an extreme NLOS environment, and the event camera's wide dynamic range is indispensable for accommodating the rapid intensity fluctuations induced by water turbulence.

3.4.4 MISO and real-time systems

Muñoz *et al.* [8] (IEEE PTL, 2026) designed a feedback-aware multiple-input single-output (MISO) event-OCC system comprising an LED array transmitter and an event-camera receiver, achieving near-real-time operation through event-driven sensing coupled with a lightweight demodulation scheme. The feedback channel enables the transmitter to adapt its modulation parameters predicated on receiver-side event-rate feedback, thereby closing the control loop for adaptive event-OCC.

Stage 4 summary. Stage 4 has brought event-OCC to deployment-relevant regimes: 320 m outdoor range, 240 kbit/s aggregate multi-node throughput, underwater operation, and closed-loop adaptation. The data-rate–versus–distance frontier has advanced by two to three orders of magnitude relative to Stage 1.

4 Modulation and Demodulation Design Space

The distinctive sensing model of the event camera—asynchronous, differential, and polarity-labeled—has engendered a modulation design space that differs fundamentally from both photodiode-based VLC and frame-based OCC. The proposed schemes can be classified into four categories.

4.1 Edge-Transition-Based Schemes

The most extensively employed approach exploits the native output of the event camera: each intensity transition generates an event. In OOK modulation, a logical "1" maps to an ON-event and a logical "0" to an OFF-event; the receiver demodulates by evaluating the polarity of the event stream within each symbol period [9, 10, 15]. The principal challenge lies in clock recovery: prolonged runs of identical bits produce no events, causing the receiver to lose symbol alignment. Two mitigation strategies have been demonstrated:

- **8b/10b line coding** [9]: guarantees a minimum of one transition every five symbols, ensuring continuous event generation at the expense of 25% coding overhead.
- **Toggle demodulation** [15]: each symbol period is expected to contain both an ON-event and an OFF-event; the demodulator alternates its decision predicated on which event type arrives first, thereby providing intrinsic noise rejection.

4.2 Pulse-Pattern Schemes

N-pulse modulation [1] encodes a symbol as a group of N pulses with a defined temporal pattern, with the pattern adapted to the bit composition of the packet. The receiver correlates the event stream against the expected patterns, exploiting the temporal structure for noise rejection. The adaptive variant [7] dynamically selects the pattern on the basis of the packet's bit distribution, optimizing the trade-off between data rate and robustness. Pulse-shaped waveform modulation [2] further sculpts the pulse waveform to exploit the sensor's high temporal resolution, enhancing resilience to noise.

4.3 Interval-and-Position Schemes

Event Interval Modulation (EIM) [14] encodes information in the time intervals between consecutive events, enabling M-ary modulation in which each interval conveys $\log_2 M$ bits. Inverse PPM [11] and FSK [11] encode information in the temporal position and frequency of event bursts, respectively. These schemes fully exploit the microsecond temporal resolution of the event camera but necessitate precise timing extraction and exhibit sensitivity to event jitter.

4.4 Spatial Encoding

The two-dimensional pixel array of the event camera affords spatial multiplexing: the dynamic QR-code approach [13] renders a spatially encoded pattern on a display, and the event camera decodes the pattern by tracking corner features and reading the spatial cells. This approach attains the highest short-range throughput (114 kbit/s) but requires a display-class transmitter and is ultimately constrained by the camera's spatial resolution. The smart visual beacon [17] employs a simpler spatial structure—a single LED or compact array—with spatial tracking tailored to robotic applications.

5 The NLOS Event-Camera Channel

NLOS event-camera communication represents the frontier that most sharply distinguishes this technology from photodiode-based VLC and frame-based OCC. The following subsections examine the channel characteristics and the technical mechanisms that render NLOS event-OCC viable.

5.1 The Reflection Channel

In NLOS-OCC, the LED transmitter illuminates a reflecting surface (wall, ground, or object), and the event camera images the surface to detect the modulated intensity variations in the reflected light. The channel is characterized by three principal impairment mechanisms:

- **Path loss**: the reflected optical power is attenuated by the product of the LED-to-surface and surface-to-camera path losses, compounded by the surface reflectance (typically 0.05–0.9, dependent on material and wavelength).
- **BRDF-induced temporal smearing**: the bidirectional reflectance distribution function of the surface redistributes the reflected light over a range of angles, attenuating the intensity at the camera and blurring the temporal modulation. Matte surfaces produce diffuse (Lambertian) reflections accompanied by severe multipath; glossy surfaces yield more specular reflections that better preserve the temporal structure of the modulation.
- **Multipath-induced inter-symbol interference**: photons traversing different reflection paths arrive at the receiver with disparate delays, engendering inter-symbol interference. Nishar *et al.* [7] empirically confirmed that larger objects and matte finishes produce higher BER due to multipath, whereas lighter-colored, glossy surfaces exhibit superior performance.

A simplified mathematical model for the NLOS-OCC channel with a patterned reflective surface was developed by He *et al.* [5], furnishing the channel-characterization foundation for subsequent NLOS event-camera research.

5.2 Advantages of Event Cameras for NLOS Communication

Three intrinsic properties of event cameras confer particular advantages for NLOS communication:

1. **Wide dynamic range** (>120 dB): the reflected signal may be 30–60 dB weaker than the direct signal. A frame-based camera with 60–70 dB dynamic range cannot simultaneously capture the bright ambient background and the faint reflected modulation without incurring saturation or quantization loss. The logarithmic photoreceptor of the event camera accommodates this dynamic range natively.
2. **Asynchronous operation**: the propagation delay of the reflected signal is *a priori* unknown and varies with geometry. Frame-based OCC necessitates frame-level synchronization, which degenerates when the delay is not an integer multiple of the frame period. Event cameras possess no frame structure; each event is timestamped independently, enabling the receiver to recover timing directly from the event stream via a digital PLL [9, 15].
3. **Motion tolerance**: in mobile NLOS scenarios—for instance, a vehicle receiving a reflected signal from a taillight while in motion—frame-based cameras suffer motion blur that smears the rolling-shutter stripes. Event cameras are inherently free of motion blur: each pixel reports the intensity change at the instant of its occurrence, irrespective of camera motion [13].

5.3 The Static-Scene Challenge

A distinctive challenge associated with event cameras is that a static scene produces no events. In LOS OCC this does not arise as a concern (the LED is modulated, ensuring that its imaging pixels consistently exhibit intensity changes); however, in NLOS OCC the reflected intensity variation may be insufficient to exceed the event threshold, a particular concern for matte surfaces. Nishar *et al.* [7] addressed this by deliberately inducing VLC-modulated intensity variations of adequate magnitude to generate events via reflection, and by tuning the event threshold to the contrast level of the reflected signal. Future investigations may explore adaptive threshold control—dynamically adjusting the per-pixel event threshold predicated on the local reflected signal strength.

6 Open Challenges and Roadmap

NLOS throughput. The current NLOS event-OCC throughput (on the order of kbit/s via passive reflection) lags the LOS frontier (80–240 kbit/s) by one to two orders of magnitude. Closing this disparity necessitates: (i) higher-gain optical front-ends (telescopic lenses, Fresnel concentrators) to collect a greater fraction of the reflected photons; (ii) adaptive event-threshold control matched to the contrast of the reflected signal; and (iii) channel coding (e.g., Raptor/RaptorQ fountain codes) tailored to the burst-error profile characteristic of the reflection channel.

NLOS channel modeling. Accurate analytical and ray-tracing models for the event-camera NLOS channel—incorporating BRDF, multipath, and the event-generation threshold—remain lacking. The simplified model of He *et al.* [5] provides a point of departure but does not account for the differential sensing model of the event camera. A unified channel model that establishes a quantitative mapping from the optical power incident on the pixel to the event-generation rate is required.

Modulation design for NLOS. The adaptive N-pulse scheme [7] remains the sole modulation formulated specifically for NLOS event-OCC. Both EIM [14] and 8b/10b-assisted OOK [9] have been validated exclusively under LOS conditions. Extending these schemes to the NLOS regime—characterized by substantially lower signal-to-noise ratios and inter-event interference induced by multipath—constitutes an open problem.

Multi-node and networking. The multi-node demonstration of Nardo *et al.* [9] (three beams, 240 kbit/s aggregate) remains the sole multi-node event-OCC result. A comprehensive networking stack—encompassing medium access control, addressing, and relaying—for event-camera-based networks has yet to be developed. The AI-PHY and relaying provisions of the IEEE 802.15.7a-2024 standard furnish a framework, but event-camera-specific adaptations are requisite.

Cross-medium and extreme environments. The underwater event-camera ISAC system of Fang *et al.* [4] represents a promising initial foray; nevertheless, the absorption and turbulence characteristics of the underwater channel are considerably more severe than those of the atmospheric channel. Space–air–ground–sea architectures integrating event-camera OCC with RF and free-space optical (FSO) communication constitute a compelling long-term vision.

Hardware co-design. The readout bottleneck identified by Nardo *et al.* [9] ($\sim$100 lines/μs) constrains the practical bandwidth to a value substantially below the per-pixel event-rate ceiling of the sensor. Sensor-level innovations—including parallel readout architectures and on-chip event processing—are requisite to unlock the full $\sim$1 MHz bandwidth.

7 Conclusion

Event cameras have evolved from a vision-sensing innovation into a communication receiver capable of 80 kbit/s at 320 m (LOS outdoor), 114 kbit/s at short range (spatial encoding), and—most significantly—kbit/s-rate communication via passive NLOS reflections from everyday objects. The technical evolution has progressed through four stages: proof-of-concept OOK (2017), spatial encoding and beacon systems (2022–2024), the first NLOS event-camera demonstrations (2025), and long-range outdoor together with specialized deployments (2025–2026). The key enablers are the event camera's microsecond temporal resolution, dynamic range exceeding 120 dB, and asynchronous, motion-blur-free operation—properties uniquely suited to the NLOS reflection channel. The principal open challenges encompass NLOS throughput enhancement,

channel modeling, NLOS-specific modulation design, multi-node networking, and hardware co-design. As event-camera hardware continues to mature and the 6G vision of integrated sensing and communication crystallizes, event-camera-based NLOS OCC is poised to become a key enabling technology for vehicular, robotic, and IoT networks operating in cluttered, occluded, and extreme environments.

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